

Liver Cirrhosis Prediction Study

Comparing Custom Neural Architectures and Statistical
Baselines

Machine Learning Research Presentation

Study Objectives

- **Clinical Outcome Tracking**
 - Deconstruct primary biliary cirrhosis biomarkers to isolate high-risk patient survival trajectories.
 - Evaluate how continuous metabolic metrics (like Bilirubin and Albumin) flag critical changes in patient status.
- **Algorithmic Engineering Comparison**
 - Perform a direct, head-to-head benchmark between a traditional parametric model and non-linear multi-layered neural networks.
 - Measure the practical performance of an identical network design implemented via manual NumPy code versus a production-grade Keras pipeline.
- **Optimization Dynamics Analysis**
 - Map how varying optimization loops navigate convex versus non-convex error surfaces over massive epoch horizons.
 - Determine if the mathematical complexity of deep hidden layers provides a tangible predictive advantage when working with localized, restricted medical data pools.

The Mayo Clinic PBC Dataset

- **Core Cohort Dimensions**

- **Source Registry:** Longitudinal study tracking 312 fully documented patient records from the Mayo Clinic trial.
- **Data Scale:** Highly focused, small-scale medical matrix featuring a deep feature-to-observation signal density.
- **Target Optimization:** Refined multi-class patient outcomes into a clean binary classification mapping ("Stable" vs. "Deceased") to maximize model stability.

- **Primary Clinical Elements**

- **Continuous Metabolic Indicators:** Tracks serum Bilirubin (mg/dl), serum Albumin (gm/dl), and Prothrombin times to measure real-time liver synthetic function.
- **Discrete Histological Markers:** Captures structural tissue degeneration grades from Stage 1 through Stage 4 cirrhosis.
- **Systemic Complication Vectors:** Integrates binary scoring records for critical physical indicators including Ascites, Edema, and Spiders.

RangeIndex: 312 entries, 0 to 311

Data columns (total 20 columns):

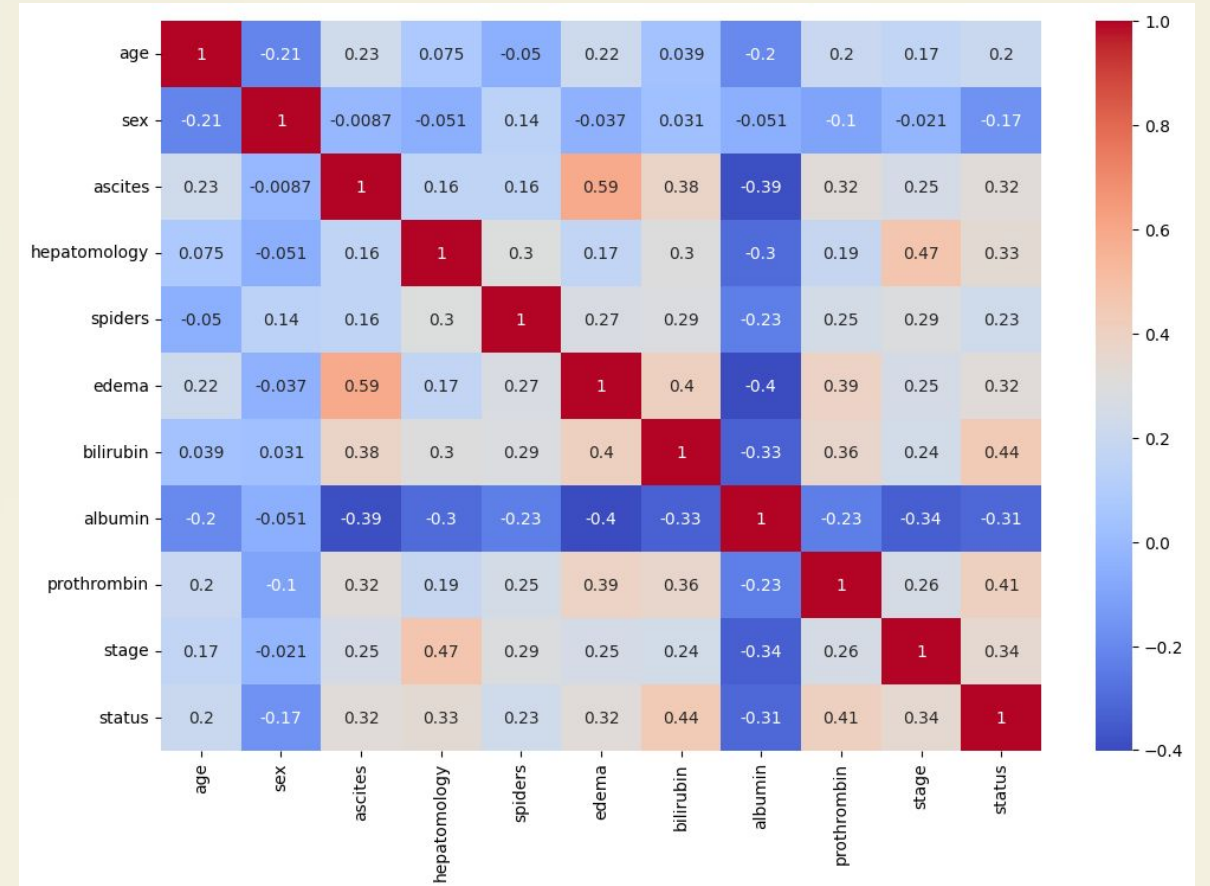
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2	status	312 non-null	int64
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9	edema	312 non-null	float64
10	bilirubin	312 non-null	float64
11	cholesterol	284 non-null	float64
12	albumin	312 non-null	float64
13	copper	310 non-null	float64
14	phosphatase	312 non-null	float64
15	SGOT	312 non-null	float64
16	triglycerides	282 non-null	float64
17	platelets	308 non-null	float64
18	prothrombin	312 non-null	float64
19	stage	312 non-null	int64

dtypes: float64(10), int64(10)

Visualizing Correlations

Feature Interdependence

- **Bilirubin:** High positive correlation with the final histological stage.
- **Albumin:** Inversely related to status, indicating serum protein decline in advanced cases.
- **Multicollinearity:** Identified between enzymatic markers (SGOT, Phosphatase), guiding feature selection.



Preprocessing Pipeline



Data Cleaning

Handling missing values and mapping categorical status [0, 1] as stable and [2] as deceased.



Encoding

One-hot encoding applied to multi-class target vectors for robust neural network classification.



Normalization

Z-score standardization used to ensure feature weights converge uniformly during backpropagation.

MVLR: The Baseline Benchmark

85.7%

MVLR Test Accuracy

Statistical Superiority

- **Mathematically Stable:** Operates on a convex error surface, guaranteeing that the model converges to the true global minimum rather than getting trapped in local minima.
- **Low Computational Demand:** Scaled efficiently and reached peak accuracy in only 4,000 epochs, a fraction of the 60,000 epochs needed for the custom ANN.
- **Clear Baseline:** Provides an interpretable, predictable benchmark to prove whether the complexity of a deep neural network adds real value.

Custom (Jaime) ANN Model Design

79.3%

ANN Test Accuracy

Architecture & Optimization

- **Topography & Architecture:**
 - Uses a vectorized $[10 \rightarrow 22 \rightarrow 11 \rightarrow 2]$ layout with Tanh hidden activations and a sigmoid output layer to handle non-linear clinical boundaries.
- **Pure NumPy Backpropagation:**
 - Engineered completely from scratch without high-level library dependencies, manually calculating local error derivatives and chain-rule matrix operations.
- **Optimization Profile:**
 - Trained over an extensive 60,000-epoch horizon to give the manual gradient descent loops ample time to navigate a complex, non-convex error surface.

Tensorflow Keras (Adam) Model Design

80.9%

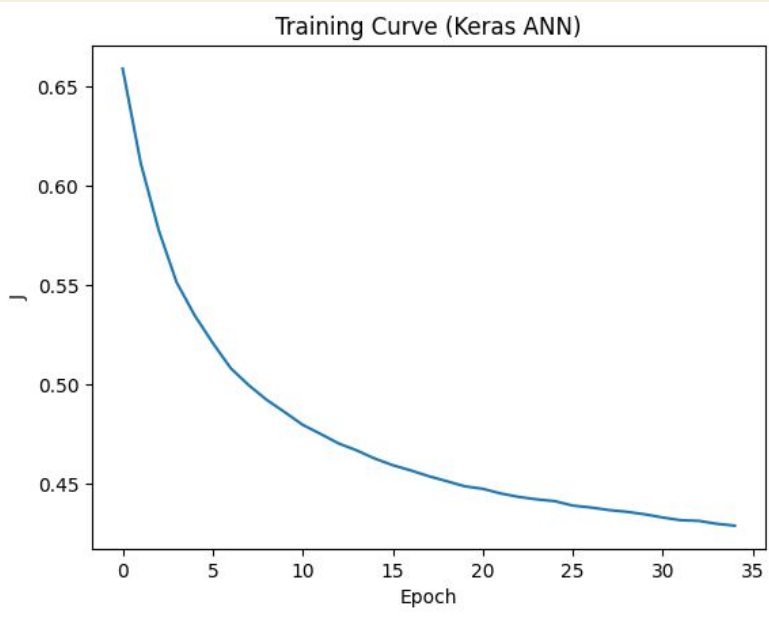
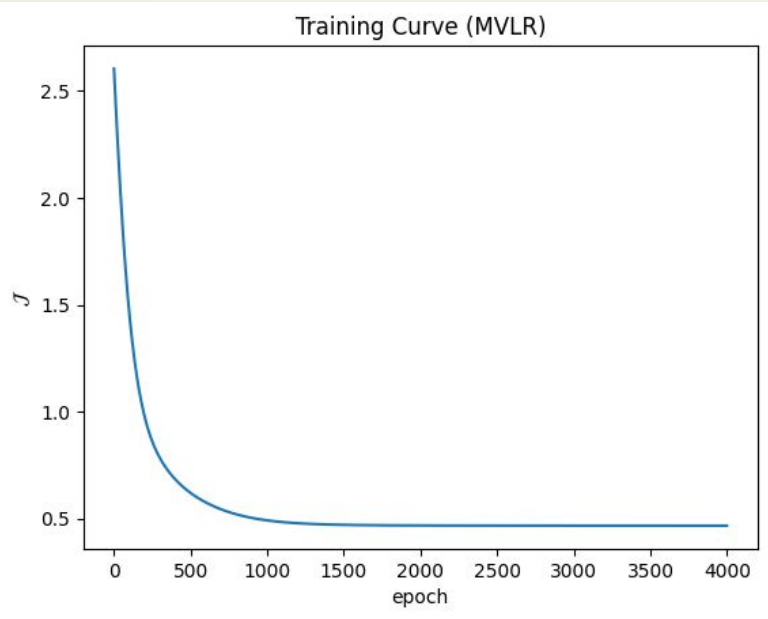
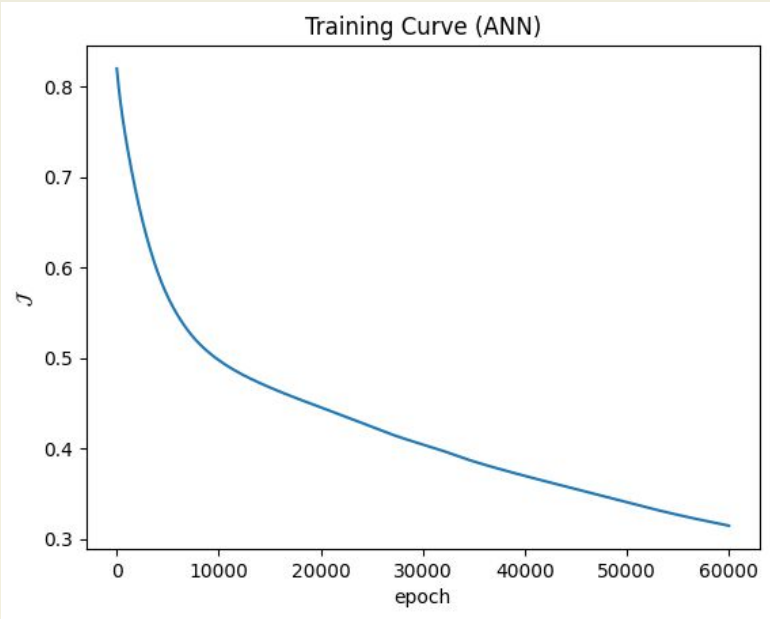
ANN Test Accuracy

Architecture & Optimization

- **Structural Control Alignment**
 - Built via the Keras Sequential API to mirror the exact layer layout and activation selections of the custom model as a strict mathematical sanity check.
- **Adaptive Momentum Optimization**
 - Compiles with the modern Adam optimizer, dynamically scaling learning rates by tracking running averages of both the first and second moments of the gradients.
- **Algorithmic Efficiency**
 - Achieved rapid weight convergence and a competitive 82.54% validation accuracy in a tiny fraction of the epochs required by the manual loops.

Performance Comparison

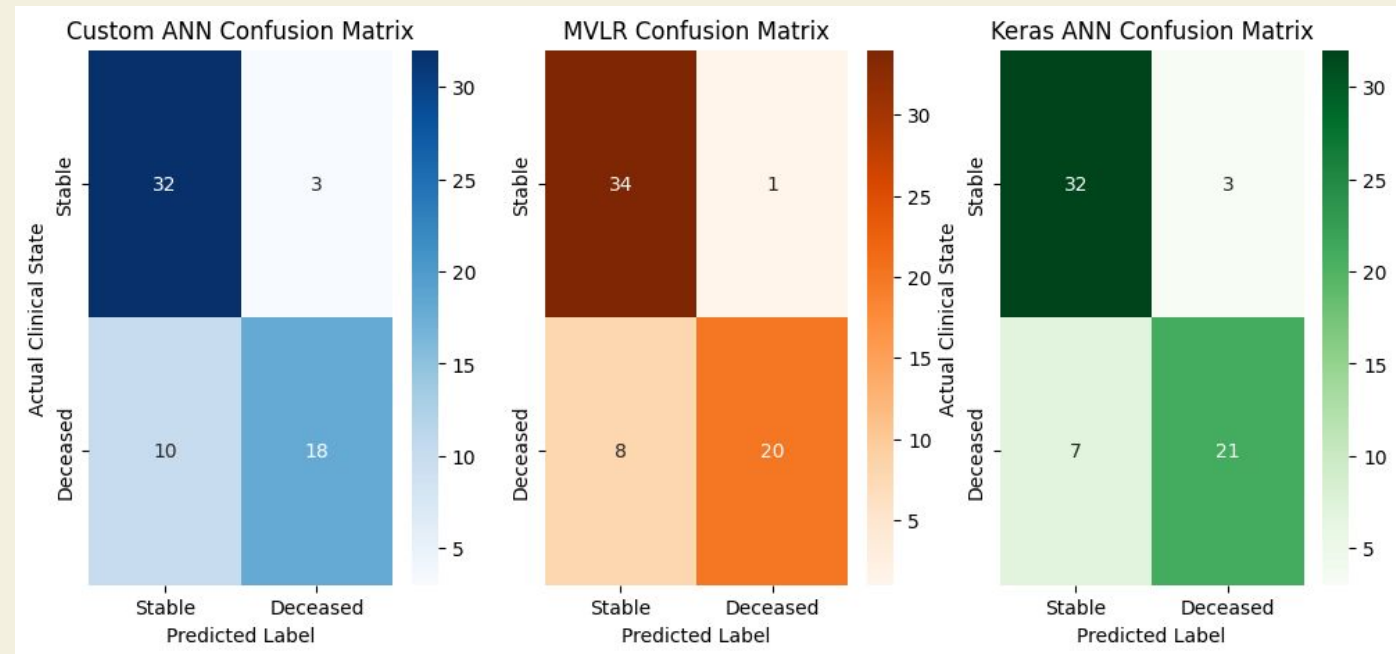
Architecture	Implementation	Epochs	Accuracy
Custom ANN	NumPy (Jaime Custom)	60,000	79.37%
MVLR	NumPy (Jaime Custom)	4,000	85.71%
Keras ANN	TensorFlow / Adam	35	84.13%



Error Analysis

Confusion Matrix Insights

- **Custom ANN:** Shows higher false negative misclassification rates, misclassifying 10 deceased profiles as Stable.
- **Keras:** Performance is much higher when compared to ANN, correctly identifying 21 – 28 deceased profile individuals.
- **Clinical Impact:** MVLR proves the most reliable model for this dataset, identifying 20–28 high-risk deceased individuals while only misclassifying a single patient.



Addressing Limitations & Future Work

Insights

- **High Validation Variance:** With a dataset of 312 records, an 80/20 split only yields 63 profiles.
 - This split introduces “data luck” and can cause model evaluation to fluctuate wildly.
- **Solution: K-Fold Cross-Validation**
 - **Performance:** Migrating to a $K = 5$ or $K = 10$ fold CV framework ensures every profile is used for both training and validation of our dataset..
 - **Mitigating Over-Fitting:** Averaging cross-entropy loss score and accuracy scores across folds reduces noise and gives true unbiased results.
 - **Data Stratification:** Using Stratified K-fold guarantees, that each fold maintains the same percentage of “Stable” versus “Deceased” patients as the total population of the dataset.
- **Complexity Paradox:**
 - Our Custom ANN and Keras did not outperform the simpler MVLR baseline. Multi-Variate Logistic Regression achieved the highest overall accuracy (85%)
 - Over-Parameterization Risk: On small datasets, adding hidden layers introduces thousands of parameters, increasing risk of the networks memorizing the localized noise instead of identifying patterns.

Questions?

Thank you for your attention.

Liver Cirrhosis Status Prediction Research
Custom Neural Network Implementation